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Lesson Notes

MODULE 6 | LESSON 3

THE HOUSING FINANCE PROBLEM AND ITS SOLUTIONS

Reading Time	30 minutes
Prior Knowledge	Publicly traded, Supply and demand
Keywords	Urbanization, Financialization, REIT, REOC, Listed, GDP

In Lesson 2, we started talking about finance as a verb: how does a real estate developer finance its development activities? Like most businesses, developers use a mix of debt and equity financing. We also discussed the primary "housing finance" decision that every household must make: to rent or buy a home. We saw how expensive buying a home is in various global locations in terms of working years and drafted a quantitative framework for making that decision assuming that a mortgage is an available financing option.

In this lesson, we again consider ways to finance housing in the absence of mortgages, that is, through various other arrangements among some combination of borrowers, renters, banks, governments, humanitarian financial organizations, and other partnerships.

1. Financing Urbanization in Africa

In Lessons 1 and 2, we discussed financing and housing from the perspective of both the homeowners or renters who demand (use and pay for) housing and the developers who supply the housing. Aside from the last lesson's discussion of

renting and developer financing, most of the discussion about housing finance so far has focused on mortgages and, by extension, mortgage-backed securities. However, mortgages from banks are by no means the only option, and in many places in the world, such mortgages are hardly an option at all. The first required reading of this lesson (Bah et al.) investigates the successes and failures of various attempts at balancing these two sides of the housing finance market in Africa, where the population has increased on average by over 2.5% annually for several decades, and this growth is expected to continue:

“Africa’s population will grow from 1.18 billion in 2015 to 2.44 billion in 2050. Although Africa’s population lives predominantly in rural areas (60 percent), the rate of rural flight to urban areas is alarming. The urbanization rate between 2000 and 2015 averaged 3.5 percent, which is the highest rate in the world” (Bah et al. 3).

The valuation or risk analysis of an investment in real estate must account for demographic projections such as these. Obviously, this massive urbanization creates a challenge in terms of housing the growing population and therefore also in terms of financing this new housing. In the author's words (Bah et al. 57):

“On the demand side, the availability of and access to housing finance is a significant determinant in a household’s decision to acquire, build, or rent a house. Similarly, on the supply side, developers need financing to build the mass housing projects that are needed to address the continent’s housing deficit.”

Bah et al. go on to provide an overview of the housing finance situation in Africa, where the “housing finance market is small and primarily caters to the needs of upper-middle- and high-income households, to the detriment of low- and middle-income families” (59). One metric used to support this statement is the level of “mortgage loans as a percentage of GDP,” where GDP is the “gross domestic product” or the monetary value of all goods and services produced in a country. Take note of whether the scatter plot in figure 3.1 (Bah et al. 60) shows any correlation between these levels and the percentage of the population that lives in urban areas, especially for the Middle East and North Africa as well as sub-Saharan Africa. Indeed, the positive correlation between mortgage loans and urbanization suggests that the increasing

urban population in these two regions has relied on mortgages to finance their housing. Nevertheless, at less than 1%, the mortgage financing of many of the countries in these regions is far below that of other developing countries. In order to meet the housing finance demand in these places, either the mortgage market must grow or a viable alternative must be found.

Bah et al. describe how several potential housing finance solutions have fared so far: each has its own risks, limitations, and constraints. Pay attention to these, as any useful model to assess the risk and value of investments in one of these programs would include these parameters. As you read about these financing alternatives, ask the following questions (Bah et al. answer them all rather explicitly):

Housing Microfinance (HMF):

- What are the likely positive outcomes when a microfinance institution begins to provide HMF in addition to its other lending activities? Note: *Microfinance institutions* (MFI) are financial institutions that provide small loans to poor households and small businesses.
- What are the main drivers of market growth for HMF?
- What are the main constraints on the growth of this financing activity?

Contractual savings for housing (CSH):

- What is the difference between an open and closed CSH system?
- In what ways does a CSH minimize credit risk?
- What are the main challenges to the viability of CSH systems?

Housing bonds:

- Based on the examples provided by Bah et al., what type of financial institution is most likely to be successful when issuing housing bonds?

Mortgage liquidity facilities (MLF):

- What are the benefits of mortgage liquidity facilities for the main stakeholders: the lenders, the homebuyer borrowers, and the government?

- In what ways do MLFs help improve standards in the lending market?

Intermediate housing loans or micromortgages:

- Why has the progress of South Africa's micromortgage program been so slow?

Integrated partnerships:

- What are the innovative ways that various partnerships are expanding access to housing finance for low- and middle-income households?
- How well does traditional mortgage underwriting assess "informal employment" when considering credit risk?
- Residential REITs have proven successful in many countries, so why are there none in Africa?

In addition to reinforcing much of what has been covered so far (in this and previous modules) with respect to credit risk analysis and housing finance (from the perspectives of the lender, the developer, and the homeowner or renter), this reading is meant to encourage the creative and constructive application of financial engineering concepts. As Bah et al. explain, "[T]o push the housing finance frontier and extend access to those in need [...] requires actions and measures within the housing finance system geared toward devising and promoting innovative business models that deal with the associated risks" (98). Financial engineering involves the techniques and understanding to meet challenges such as this with robust and innovative solutions grounded in both state-of-the-art theory and up-to-date empirical research. With that said, the next section of this lesson (and the reading mentioned there) touches on certain side effects of the expansion of finance into more and more aspects of our lives in modern society.

2. REITs, REOCs, and Renting

The second required reading further investigates the financing, and the financialization, of the rental housing market, especially the role of REITs and REOCs (real estate operating companies) in both of these phenomena. What is financialization? Fuller reiterates Aalbers's definition of the term as the most important and appropriate:

“[Financialization means] the increasing dominance of financial actors, markets, practices, measurements and narratives, at various scales, resulting in a structural transformation of economies, firms (including financial institutions), states and households.... [F]inancialization means the increasing role of financial motives, financial markets, financial actors, and financial institutions in the operation of the domestic and international economies” (552).

Based on this description, the concept has clear significance for almost every topic in financial engineering, including (obviously) housing finance.

In Module 5, we discussed the collapse of the mortgage market, which brought on the Great Financial Crisis. Fuller mentions that REITs and REOCs bought up many of the real estate portfolios that were sold off at this time (555), which explains why they are among “the largest landlords in the U.S. and Canada—and Germany-based REOC Vonovia is the single largest corporate landlord in Europe, owning over 400,000 homes” (Fuller 556). REITs and REOCs are structurally very similar (Fuller 555). Both REITs and REOCs:

- Primarily own residential real estate
- Derive most of their income from rent or property sale
- Are publicly listed on an exchange
- Eliminate the hassle of actually owning, letting, and maintaining physical property, and
- Allow even small investors to buy a slice of the income generated by a number of different properties.

Figure 1: REITs vs. REOCs

REITs	REOCs
“Pass-through” entities to distribute the vast majority of their profits to shareholders in the form of dividends	Ordinary corporations that more easily reinvest profits into their operations
Shareholders pay the required taxes	Pay corporate tax on their own earnings

Adapted from: Fuller, Gregory W. "The Financialization of Rented Homes: Continuity and Change in Housing Financialization." *Review of Evolutionary Political Economy*, vol. 2, 2021, p. 555. <https://doi.org/10.1007/s43253-021-00050-7>.

As you read about how REITs developed across regions, think about the factors you would want to be sure to include in a pricing or risk model for an REIT investment in the United States, Europe, and East and Southeast Asia. For example, in

the U.S., if a particular REIT indicates that its properties are located near employers such as Amazon, Facebook, and Google, perhaps you would want to include information from forecasts relating to these companies. If Google announces plans for expansion or cost cutting, there may be an impact on the performance of certain REITs. Regulation and politics also have price and risk implications; for example, statutory rent controls would certainly impact profits from residential rental properties. So does litigation: if an REIT's property management is overly aggressive with its tenants, a class-action lawsuit could mean significant losses. Finally, pay attention to the three ways that Fuller claims REITs and the financialization of rental housing may benefit the housing market (related to incentives, transparency, and the alignment of interests).

3. Conclusion

This lesson introduced various novel ways that governments and financial markets have sought to address or profit from (respectively) the housing situation. In most cases, many people still do not have access to housing finance. We discussed how REITs and REOCs are shaping the housing market, and we discussed many factors to consider when analyzing an REIT, such as regulations, demographics, economic conditions, and even how the performance of certain technology companies can affect prices for certain REITs.

In the next lesson, we discuss this type of fundamental analysis in more detail, and we compare it to other strategies for selecting investment opportunities, especially stocks.

References

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